

Reproduction in Organisms

Class 12 Biology NCERT

Introduction to Reproduction

Reproduction is a fundamental biological process by which organisms produce young ones (offspring) similar to themselves. It enables the continuity of species, generation after generation. Every individual organism has a specific **life span**, which is the period from birth to its natural death.

- Reproduction is essential for the continuation of species.
- Life span varies greatly among different organisms.
- No individual is immortal, except single-celled organisms.

Asexual Reproduction

Asexual reproduction is a mode of reproduction in which a single parent produces offspring that are genetically identical to the parent. The offspring are referred to as **clones**, meaning they are exact copies both morphologically and genetically.

- Single parent involved.
- Offspring are genetically identical to the parent.
- Common in single-celled organisms, plants, and simple animals.

Modes of Asexual Reproduction

Different organisms employ various methods for asexual reproduction.

Binary Fission

In **binary fission**, the parent cell divides into two halves, each of which rapidly grows into an adult. This method is common in unicellular organisms.

- Parent cell divides into two equal daughter cells.

- Example: *Amoeba*, *Paramecium*.

Budding

Budding involves the formation of a small outgrowth or bud from the parent body, which eventually detaches and develops into a new individual.

- Unequal division, forming a bud.
- Example: Yeast, *Hydra*.

Spore Formation

Many simple organisms reproduce through specialized spore structures.

- **Zoospores:** Microscopic, motile asexual spores (e.g., *Chlamydomonas*).
- **Conidia:** Non-motile spores produced externally (e.g., *Penicillium*).
- **Gemmules:** Internal buds found in sponges (e.g., Sponges).

Vegetative Propagation

In plants, asexual reproduction is often called **vegetative propagation**. Specialized structures called **vegetative propagules** are involved in this process.

- **Runner:** Grass, *Oxalis*
- **Rhizome:** Ginger, Turmeric, Banana
- **Tuber:** Potato (eyes are axillary buds)

More Vegetative Propagules

Additional vegetative propagules include:

- **Offset:** Water hyacinth (*Eichhornia*), *Pistia*
- **Bulb:** Onion
- **Bulbil:** *Agave*

- **Leaf Buds:** *Bryophyllum* (adventitious buds on leaf margins)

Water Hyacinth: Terror of Bengal

The water hyacinth (*Eichhornia crassipes*) is an invasive aquatic weed. It spreads very rapidly through **offset** vegetative propagation. Its immense growth drains oxygen from the water, leading to the death of fish and other aquatic organisms. This has earned it the name 'Terror of Bengal'.

- Invasive aquatic weed.
- Rapid spread via offset.
- Depletes oxygen, harming aquatic life.

Sexual Reproduction

Sexual reproduction involves the formation of male and female gametes, either by the same individual or by different individuals of the opposite sex. These gametes then fuse to form a zygote, which develops into a new organism.

- Involves formation and fusion of gametes.
- Offspring are not identical to parents (variation).
- A more complex and slower process.

Phases in Life Cycle

Organisms undergo different phases in their life cycle.

Juvenile Phase (Vegetative Phase)

This is the period of growth before an organism can reproduce sexually. In plants, it is known as the **vegetative phase**.

- Period of growth before reproduction.
- Called vegetative phase in plants.

Reproductive Phase

The period during which an organism is capable of sexual reproduction. In higher plants, this is marked by **flowering**.

- Animals show distinct reproductive cycles.
- **Oestrous cycle**: Non-primate mammals (e.g., cows, sheep, rats).
- **Menstrual cycle**: Primates (e.g., monkeys, apes, humans).

Senescent Phase

The end of the reproductive phase is marked by the **senescent phase**. This is characterized by a gradual deterioration in body functions, leading to old age and ultimately death.

- End of reproductive phase.
- Decline in metabolic activity.
- Leads to death.



Events in Sexual Reproduction

Sexual reproduction involves a sequence of events: pre-fertilisation, fertilisation, and post-fertilisation.

Pre-fertilisation Events

These are all the events in sexual reproduction that occur prior to the fusion of gametes.

- Includes **gametogenesis** (gamete formation) and **gamete transfer**.

Gametogenesis

The process of formation of haploid male and female gametes. Gametes are always haploid (n).

- **Homogametes (Isogametes)**: Morphologically similar gametes (e.g., some algae).

- **Heterogametes:** Morphologically distinct gametes (e.g., most organisms).
- Male gamete: **Antherozoid** or **sperm**.
- Female gamete: **Ovum** or **egg**.

Sexuality in Organisms

Organisms can be classified based on the presence of reproductive structures.

- **Monoecious/Homothallic:** Both male and female structures on the same individual (e.g., *Cucurbits*, coconut, *Chara*).
- **Dioecious/Heterothallic:** Male and female structures on different individuals (e.g., Papaya, date palm, *Marchantia*).
- **Bisexual animals (Hermaphrodites):** Possess both male and female organs (e.g., Earthworm).
- **Unisexual animals:** Possess either male or female organs (e.g., cockroach, humans).

Cell Division During Gamete Formation

The type of cell division for gamete formation depends on the parent organism's ploidy.

- **Diploid (2n) parent:** Undergoes **meiosis** to produce haploid gametes (e.g., humans, higher plants).
- **Haploid (n) parent:** Undergoes **mitosis** to produce haploid gametes (e.g., algae, fungi, bryophytes).

Gamete Transfer

After formation, gametes must be physically brought together for fusion. Male gametes are typically motile, and female gametes are stationary.

- **Water** is a common medium for gamete transfer in simple plants (algae, bryophytes, pteridophytes).
- In seed plants, **pollen grains** carry male gametes to the stigma (pollination).

- Pollination can be **self-pollination** or **cross-pollination**.

Fertilisation (Syngamy)

Fertilisation or **syngamy** is the process of fusion of male and female gametes, resulting in the formation of a diploid **zygote** ($2n$).

- **Parthenogenesis**: Development of new organism without fertilisation (e.g., rotifers, drones of honeybees).

Types of Fertilisation

Fertilisation can occur either outside or inside the organism's body.

- **External fertilisation**: Occurs outside the body, in an external medium (e.g., fish, frogs, most algae).
- **Internal fertilisation**: Occurs inside the body of the female organism (e.g., reptiles, birds, mammals, most plants).



Post-fertilisation Events

These are the events that occur after the formation of the zygote.

- Includes **zygote formation** and **embryogenesis**.

Zygote Development

The zygote is the vital link that ensures continuity of species between generations. Its development depends on the life cycle of the organism and the environment.

- In algae/fungi (haplontic life cycle), zygote undergoes meiosis to form haploid spores.
- In diplontic/haplodiplontic life cycles, zygote undergoes mitosis.

Embryogenesis

The process of development of the embryo from the zygote. It involves cell division (mitosis) and cell differentiation.

- **Cell division:** Increases cell number.
- **Cell differentiation:** Cells modify to form tissues and organs.

Development in Animals

Animals are classified based on where zygote development occurs.

- **Oviparous animals:** Lay eggs; development occurs outside the body (e.g., reptiles, birds).
- **Viviparous animals:** Give birth to live young; development occurs inside the female body (e.g., most mammals).
- Viviparity offers better embryonic care and protection.

Development in Flowering Plants

In flowering plants, post-fertilisation changes lead to seed and fruit formation.

- Zygote → Embryo (inside ovule).
- Ovules → Seeds.
- Ovary → Fruit (with protective pericarp).